MILITARY SPECIFICATION
CARTRIDGE, DELAY: XM271, ASSEMBLY

1. SCOPE

1.1 This specification covers Cartridge, Delay: XM271, Assembly, for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-H-25579	- Hose Assembly, Tetrafluorethylene, High Temperature, Medium Pressure
MIL-C-60539	- Coating, Anodic, Conventional, for Aluminum and Aluminum Alloys

STANDARDS

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-rings
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	- Ammunition Data Card

DRAWINGS
Munitions Command

D10551590	- Cartridge, Delay: XM271, Assembly
C8864168	- Carton, Inner Packing Ammunition for Cartridge Initiator Delay M84, M85, M86, M87, M88, M89, M165, XM128, XM148, XM159, M174 or M175

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C9214156 - Box, Packing, Ammunition for Cartridge Initiator Delay M84, M85, M86, M87, M88, M89, M165, XM128, XM148, XM159, M174 or M175

IEL-10551590 - Inspection Equipment List for Cartridge, Delay: XM271, Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies, shall conform to the applicable drawings and referenced specifications.

3.2 First article sample.- Requirements for the submission of first article samples by contractors shall be as specified in the contract.

3.3 Protective finish.- Protective finish shall be in accordance with MIL-C-60539.

3.4 Vibration.- The cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant or derangement of components.

3.5 Functional.

3.5.1 The cartridge shall not fail to fire.

3.5.2 Ballistics.- The cartridge when tested in an XM110 Initiator shall comply with the requirements specified in table I.

Table I

Temperature	Pressure (psi) 1/ Min	Max	Delay Time, Seconds (See 6.1)
-65° to -70° F	800	1350	1.6 max
65° to 75° F	1000	1400	1.4 ± 0.1
200° to 205° F	900	1500	1.1 min

1/ Pressure values shall be produced in a one cubic inch pressure station at the end of 15 feet of hose (Size No. 4) in accordance with MIL-H-25579.

3.6 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

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3.6.1 Metal defects.- The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.6.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 23 anodized cartridge cases and 23 anodized retaining rings for protective finish tests, 50 cartridge cases for hardness tests, 12 cartridges for vibration tests (to be selected from ballistic samples), 5 cartridges for sealant tests and 30 cartridges for ballistic tests and tested in accordance with table II and 10 sets of unassembled components (including explosive components) for dimensional and visual examination.

Table II

Paragraph Requirement	Paragraph Test	Test	Sample Size	Type of Test: Destructive	Type of Test: Nondestructive
3.3	4.4.1	Protective finish	23 Cartridge Cases	X	—
			23 Retaining Rings	X	—
Dwg	4.4.2	Hardness	50 Cartridge Cases	X	—
Dwg	4.4.3	Airtightness	Each	—	X
3.4	4.4.4	Vibration	12 (from Ballistic Samples)	—	X
Dwg	4.4.5	Sealant application	5 Cartridges	X	—
3.5	4.4.6.1	Ballistics	30	X	—

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The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests. - After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics. - The sample for ballistic tests shall be temperature conditioned and tested to determine compliance with table I. The sample for testing shall consist of the following:

Table III

Temperature	Vibrated	Not Vibrated
-65° to -70° F	4	6
65° to 75° F	4	6
200° to 205° F	4	6

4.2.1.3 Initial production sample failure. - Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product. - The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot. - Unless otherwise specified by the procuring agency, an ammunition lot shall consist of 1200 cartridges, maximum, exclusive of test samples and shall consist of:

a. Components produced by a homogeneous process.

b. Propellant from not more than one lot.

c. Primers from not more than one lot.

d. Igniters from not more than one lot.

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4.3.2 Examination. - One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in tables IV and V. For an individual defect (single characteristic), the criteria in table IV shall be used. For total defects within a major or minor class, the criteria in table V shall be used. All nonconforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Inspection for rubber o-rings shall be in accordance with MIL-STD-413. Four power magnification (4X) may be used in performing visual examination for such defects as burr and foreign matter.

Table IV

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

Table V

Total Allowable Defects per Sample Size

Sample Size for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Size for Minor Defects	**110**	**150**	**225**	**300**
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

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4.3.2.1 Delay Element, XM69, Drawing D10551589.

Defect		Method of Inspection
CRITICAL		
1.	Metal defective	Visual
2.	Mixed ammunition	Visual
3.	Flash hole missing or obstructed	Visual
4.	Marking missing, incorrect or illegible	Visual
5.	Delay charge not visible through flash hole	Visual
MAJOR		
101.	Protective finish inadequate or defective	Visual
102.	Foreign matter	Visual
MINOR		
None defined		

4.3.2.2 Ring, Retaining, Drawing B8597235, covering a detail of Drawing D10551590.

CRITICAL		
None defined		
MAJOR		
101.	Concentricity of inside diameter with outside diameter	Gage
102.	Diameter inside, max	Gage
103.	Diameter outside, min	Gage
104.	Thickness	Gage
105.	Metal defective	Visual
106.	Protective finish inadequate or defective	Visual
107.	Burr	Visual
108.	Foreign matter	Visual
MINOR		
201.	Diameter inside, min	Gage
202.	Diameter outside, max	Gage
203.	Surface finish improper	Visual

4.3.2.3 Case, Cartridge, Drawing C10519996, covering a detail of Drawing D10551590.

CRITICAL		
1.	Metal defective	Visual

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Defect	Method of Inspection
MAJOR	
101. Concentricity of head with body	Gage
102. Diameter of head, min	Gage
103. Thickness of wall, min	Gage
104. Thickness of base, min	Gage
105. Protective finish inadequate or defective	Visual
106. Burr or sharp corner	Visual
107. Foreign matter	Visual
MINOR	
201. Diameter of head, max	Gage
202. Diameter of body	Gage
203. Total length	Gage
204. Thickness of wall, max	Gage
205. Thickness of base, max	Gage
206. Location of datum diameter	Gage
207. Radius missing or incorrect	Visual
208. Surface finish improper	Visual

4.3.2.4 Assembly, Cartridge Delay, XM271, Drawing D10551590.

CRITICAL	
1. Any part missing	X-ray 1/ and Visual
2. Improper assembly	X-ray 1/ and Visual
3. Metal defective	Visual
4. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Diameter of crimp, min	Gage
103. Location of crimp with delay element	Gage
MINOR	
201. Diameter of body, max	Gage
202. Diameter of head, max	Gage
203. Total length, max	Gage
204. Location of datum diameter	Gage
205. Varnish missing	Visual

1/ The cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring agency.

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4.3.2.5 Carton, Inner Packing, Drawing C8864168.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Manual and Visual
202.	Quantity of contents incorrect	Visual
203.	Separator or filler missing	Visual
204.	Heat sealer incomplete, loose or distorted	Visual
205.	Cut or damaged through impregnated box	Visual
206.	Marking missing, incorrect or illegible	Visual

4.3.2.6 Box, Packing, Drawing C9214156.

CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Manual and Visual
202.	Hardware missing, loose, broken or improperly engaged	Visual
203.	Strapping missing, loose, broken or improperly assembled	Visual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual
207.	Number of cartons incorrect	Visual
208.	Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4:

4.3.3.1 Protective finish.- Twenty-three cartridge cases and 23 retaining rings shall be selected from each anodized lot for protective finish tests. Failure of any component to comply with the requirements of 3.3 shall be cause for rejection of the lot.

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4.3.3.2 Hardness. - Fifty cartridge cases shall be selected for this test. Failure of any case to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.3 Airtightness. - Each cartridge shall be subjected to the airtightness test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the cartridge.

4.3.3.4 Vibration. - Twelve of the 30 assembled x-rayed cartridges of the initial production sample selected for ballistic tests and properly identified shall be subjected to vibration as in 3.4. After the vibration test, the 12 vibrated cartridges shall be x-rayed as directed in Note 1/, 4.3.2.4. Evidence of any disintegration of the primer, propellant, igniter, delay charge or derangement of any component shall be cause for rejection of the lot as indicated in 3.4. When a change occurs in the lot of primers, propellant, igniters or delay charge after the initiation of production, 12 assembled cartridges from the first production lot shall be tested in accordance with this paragraph.

4.3.3.5 Sealant application. - Five cartridges from each production lot shall be utilized for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.6 Functional. - The following tests shall be performed by the Government:

4.3.3.6.1 Ballistics. - The 30 assembled cartridges from each production lot shall be grouped and temperature conditioned in accordance with 4.4.6.1.1 and table VI. Failure of any cartridge to comply with the requirements of 3.5 shall be cause for rejection of the lot.

Table VI

Temperature	Sample Size
-65° to -70° F	10
65° to 75° F	10
200° to 205° F	10

4.3.4 Inspection equipment. - The examination and tests shall be performed using equipment listed on IEL-10551590.

4.4 Test methods and procedures.

4.4.1 Protective finish. - Test procedures for protective finish shall be in accordance with MIL-C-60539. All anodized components shall be sectioned longitudinally to better expose the critical surfaces to the salt spray.

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4.4.2 Hardness.- The hardness test shall be performed on all heat-treated components prior to the application of protective finish. A hardness testing machine shall be used to determine compliance with the drawing requirements.

4.4.3 Airtightness.- The method for determining airtightness shall be by use of the equipment listed on IEL-10551590.

4.4.4 Vibration.- The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.5 Sealant application.- The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length (360°) of the joining surfaces at the junction of the o-ring and case sidewall and that the propellant cavity is free of sealant.

4.4.6 Functional.

4.4.6.1 Ballistics.- Cartridges of the sample shall be grouped and temperature conditioned in accordance with 4.4.6.1.1 and rigidly supported in the test fixture. The sample shall be tested within five minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be re-conditioned in accordance with 4.4.6.1.1.

4.4.6.1.1 Temperature conditioning.- Cartridges shall be loaded into acceptable XM110 Initiators with end caps to cover the gas ports during the conditioning period. The temperature of the propellant within the cartridge shall be considered as the conditioning temperature and will be determined by means of a thermocoupler attached to the center of the propellant charge of a specimen cartridge assembled in a standard test device similar to the initiators undergoing tests. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the specimen cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8864168 and C9214156, except that all markings shall refer to Cartridge, Delay: XM271, Assembly.

5.2 Data cards.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Delay time is defined as the time that elapses between firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

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Preparing activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

DIRECT GOVERNMENT CONTRACT SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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